Full Pivoting and Ill-Conditioned Matrices

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1 Why is full pivoting necessary for the prescribed real matrix?

Full pivoting is necessary for the prescribed matrix because:

- the matrix has exact zeros on the diagonal at rows 4, 8 and 9. Without pivoting the algorithm would attempt to divide by zero at these positions, causing the program to crash entirely.
- even where diagonal entries are non-zero, they are small. Dividing by small numbers would amplify rounding errors, corrupting the result.

2 At approximately which dimension does the Hilbert matrix become numerically problematic?

The Hilbert matrix becomes unreliable around n= 13. Because the rows are nearly linearly dependent, condition number grows exponentially, rounding errors eventually crashes the solution.

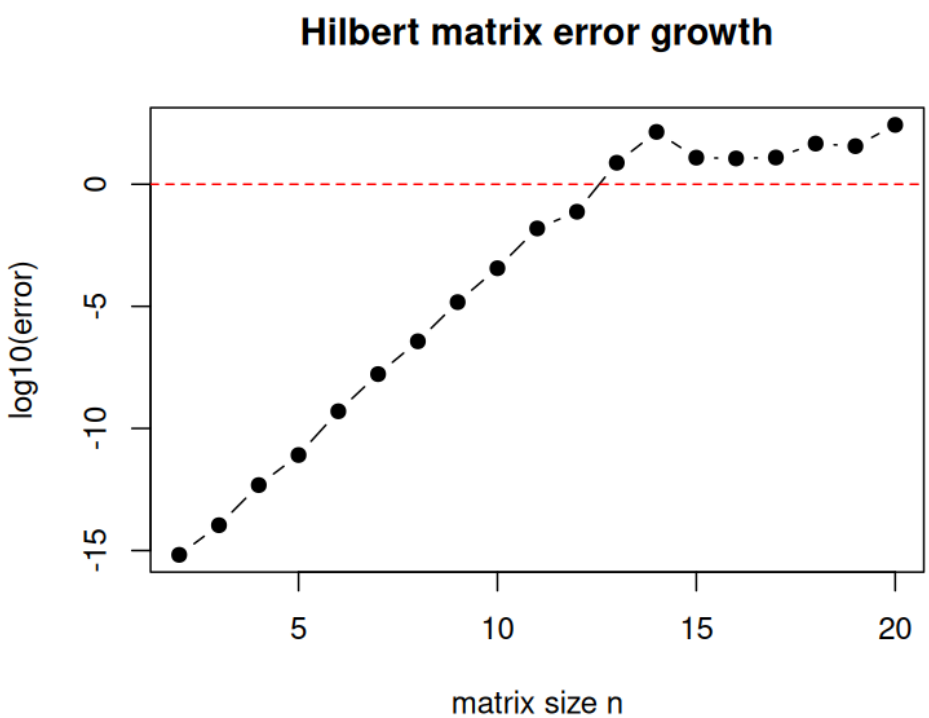

Figure 1: Error growth for the Hilbert matrix

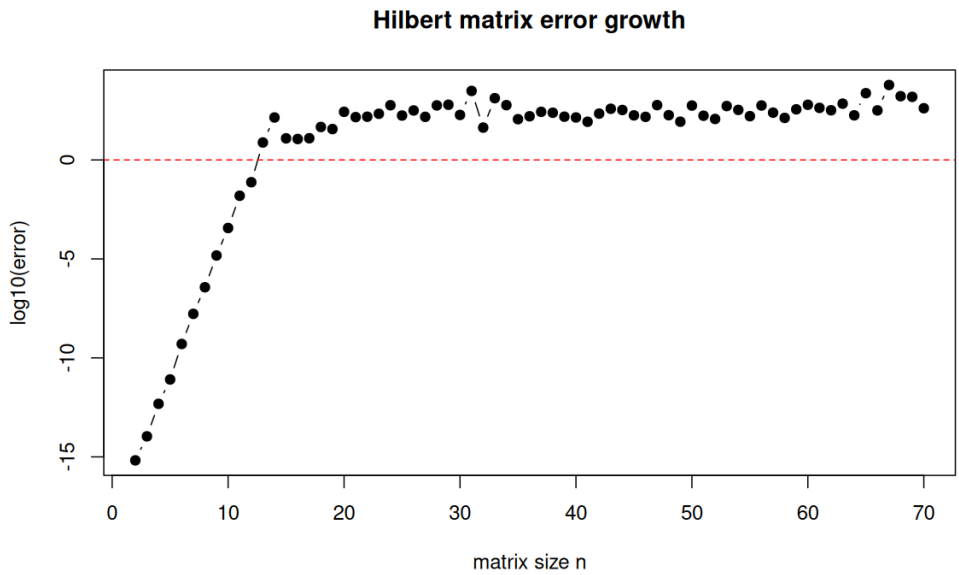

Figure 2: Error growth for the Hilbert matrix

Does the generalized Hilbert-type matrix behave similarly to the Hilbert matrix?

The generalised Hilbert-type matrix with coefficients $c = 0.83$ and $d = 1.27$, defined by

$$a_{ij} = \frac{1}{ci + dj - j}$$

behaves similarly to the standard Hilbert matrix, with a few differences.

- First, the error growth before failure is slightly less steep than the standard Hilbert matrix, suggesting that the coefficients $c = 0.83$ and $d = 1.27$ produce a somewhat better-conditioned matrix at small dimensions.
- Second, after the breakdown at $n = 13$, the error plateaus at around 10^2 rather than continuing to grow, which differs from the more erratic behaviour seen in the standard Hilbert matrix at larger dimensions.

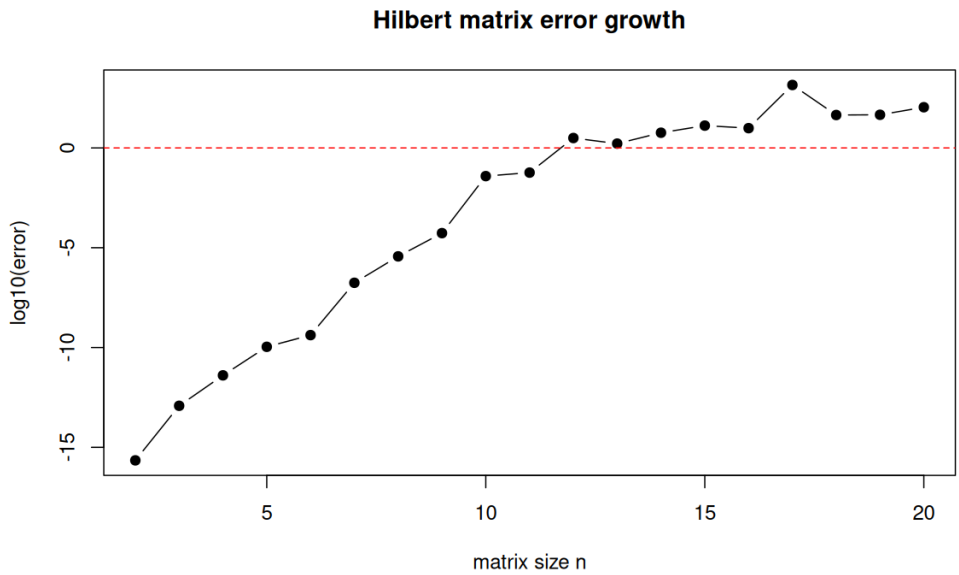

Figure 3: Error growth for the generalised Hilbert matrix n = 20, c=0.83, d=1.27

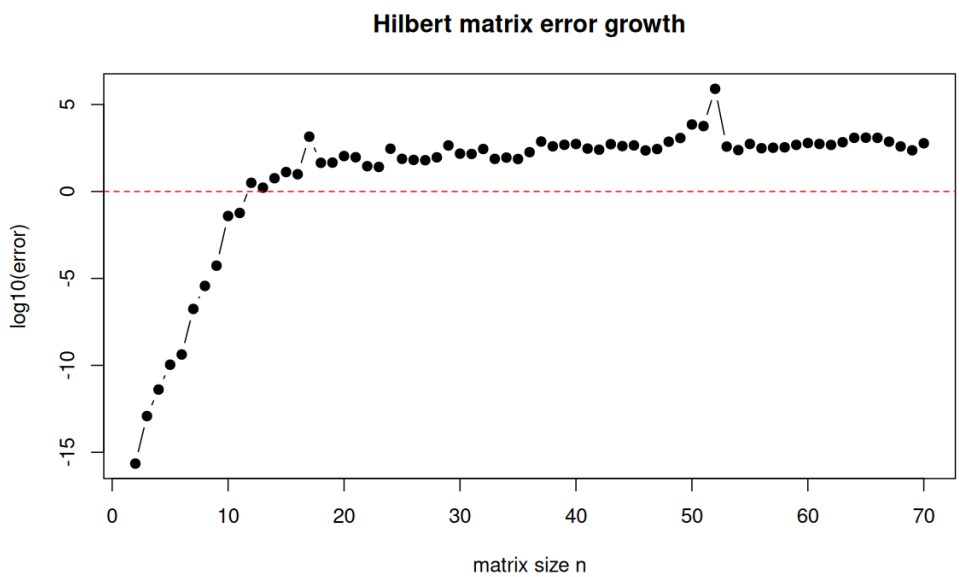

Figure 4: Error growth for the generalised Hilbert matrix n= 70, c=0.83, d=1.27

3 Is the failure sudden, or does the error grow gradually with dimension?

In the graphs, it is visible that the rounding errors are gradual at first, suggesting slow accumulation as the condition number grows. The error remains below 1.0 up to $n = 12$, meaning the solver is still producing a trustworthy result. However, after $n = 13$, the error drastically increases and the solution loses its numerical stability and even full pivoting cannot compensate for the accumulated error.